INDIAN PETROL PUMP BILLING SYSTEM USING C++

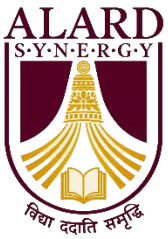
A PROJECT SYNOPSIS

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A.Y. 2026-27

Project Details

Title

Indian Petrol Pump Billing System Using C++

Introduction

The Indian Petrol Pump Billing System is a C++-based software simulation designed to demonstrate the basic billing operations performed at a petrol pump. Petrol pumps handle various activities, including fuel selection, quantity measurement, price calculation, payment collection, and receipt generation. Performing these operations accurately is important for maintaining proper transaction records and providing efficient customer service.

This project simulates the billing process through a console-based application. It allows the user to select petrol or diesel, enter the required quantity in litres, calculate the total cost based on the configured price per litre, enter the payment amount, and calculate the remaining balance or change. The system also displays a simple bill containing the fuel type, price, quantity, and total amount.

The project uses fundamental C++ programming concepts such as variables, structures, conditional statements, loops, arithmetic operations, input validation, and formatted output. It is intended for academic learning and demonstration of a basic fuel-billing workflow. The application does not control or communicate with actual petrol pump equipment.

Objectives and Scope

Objectives

The main objectives of the Indian Petrol Pump Billing System are:

1. To develop a simple petrol pump billing simulation using C++ programming.
2. To provide options for selecting petrol or diesel.
3. To calculate the total fuel cost based on the quantity entered and the configured price per litre.
4. To accept the customer's payment amount and calculate the change returned or the remaining balance.
5. To generate a simple bill displaying the fuel type, quantity, price per litre, and total amount.
6. To validate user inputs and handle invalid fuel selections, quantities, and payment values.
7. To demonstrate the practical application of basic C++ programming concepts in a real-world business scenario.
8. To understand the importance of accurate billing, transaction handling, and safety in petrol pump operations.

Scope

The scope of this project is limited to a console-based billing simulation for educational purposes. It covers fuel selection, quantity input, amount calculation, payment processing, basic validation, and receipt display.

The system uses predefined sample prices for petrol and diesel, which can be changed for classroom demonstrations. It does not store transaction history, connect to payment gateways, calculate actual taxes or accounting entries, or interface with real fuel dispensers.

The project can be extended in the future by adding transaction storage, daily sales reports, staff authentication, transaction IDs, timestamps, and a graphical user interface.

Benefits to the End-User

- Simplifies the demonstration of basic fuel-billing operations.
- Reduces manual calculation effort within the simulated billing process.
- Helps users understand how fuel quantity and price determine the total bill.
- Provides a clear and organized receipt format.
- Improves understanding of input validation and payment calculations.
- Serves as an educational foundation for developing more advanced billing applications.

Abstract

The Indian Petrol Pump Billing System is a C++ mini project developed to simulate the basic billing operations of an Indian petrol pump. The project aims to demonstrate how a computerized application can simplify fuel-cost calculations, payment handling, and receipt generation. The system provides options for selecting petrol or diesel and allows the user to enter the required quantity of fuel in litres. Based on the configured price per litre, the application calculates the total amount payable. It then accepts the payment amount and determines whether additional payment is required or change should be returned. Finally, a simple receipt is displayed with the transaction details.

The system is implemented using fundamental C++ programming concepts, including structures, variables, conditional statements, loops, arithmetic operations, input validation, and formatted output. Basic testing can be performed using valid and invalid fuel selections, positive and negative quantities, exact payments, overpayments, and insufficient payments. The project demonstrates the usefulness of programming in automating simple billing calculations and improving the clarity of transaction information. However, it is an educational simulation that uses illustrative fuel prices and does not interact with real fuel-dispensing equipment. Future enhancements may include transaction record storage, daily sales reports, staff login, unique transaction IDs, timestamps, and a graphical user interface.

Keywords: C++, Petrol Pump, Fuel Billing System, Billing Automation, Payment Processing, Receipt Generation, Input Validation.

Project Category

Main Category: 2. Application Development

Sub-Category: Console-Based Application Development using C++

Process Description

The project Indian Petrol Pump Billing System falls under Application Development because it involves developing a software application using C++ to simulate the billing operations of a petrol pump. The system allows users to select petrol or diesel, enter the fuel quantity, calculate the total bill, process payment amounts, calculate the balance or change, and generate a simple receipt. It demonstrates the practical application of programming concepts to automate basic billing calculations and improve transaction accuracy.

The project is designed for educational purposes and does not connect to or control actual petrol pump equipment.

Requirement of Resources and Limitations

1. Requirement of Resources

The Indian Petrol Pump Billing System is a console-based application developed using C++. The project requires basic computer hardware, software development tools, and sample fuel-price data to design, develop, execute, and test the proposed system.

A. Hardware Requirements

- **Computer/Laptop:** A basic computer or laptop capable of running a C++ compiler.
- **Processor:** Any processor that supports the selected operating system and development environment.
- **RAM:** Minimum 2 GB RAM recommended.
- **Storage:** A small amount of free disk space for the source code, compiler, and project documentation.
- **Input/Output Devices:** Keyboard for entering fuel details and payment amounts, and a monitor for displaying billing information and receipts.

B. Software Requirements

- **Programming Language:** C++.
- **Compiler:** GCC/G++, MinGW, or another compatible C++ compiler.
- **Development Environment:** Visual Studio Code, Code::Blocks, Dev-C++, or any suitable C++ IDE.
- **Operating System:** Windows, Linux, or another operating system supported by the chosen compiler.
- **Libraries:** Standard C++ libraries such as iostream, iomanip, string, and limits.
- **Testing Tools:** Compiler diagnostics and manual test cases for checking fuel selection, quantity input, bill calculation, payment processing, and receipt generation.

C. Data Requirements

- Fuel types, such as petrol and diesel.
- Configured sample prices per litre.
- Fuel quantity entered by the user.
- Payment amount entered by the user.
- Calculated bill total, outstanding balance, or change returned.

The fuel prices used in the program are illustrative values and should be updated when required for classroom demonstrations.

D. Human Resources

- **Project Developer:** Responsible for designing, coding, testing, and documenting the application.
- **Project Guide/Supervisor:** Provides technical guidance, reviews the implementation, and evaluates the project.
- **End-User/Tester:** Enters sample transaction details and verifies the billing calculations and displayed output.

2. Limitations of the Proposed System

The proposed system demonstrates the basic billing workflow of a petrol pump. However, compared with a comprehensive commercial petrol pump management system, it has the following limitations:

1. **Console-Based Interface:** The system operates through a text-based interface and does not provide a graphical user interface or touchscreen functionality.
2. **Fixed Fuel Prices:** The program uses predefined sample prices for petrol and diesel. It does not automatically retrieve or update actual fuel prices.
3. **No Transaction Database:** The system does not permanently store transaction records, customer details, or historical billing information.
4. **No Real Payment Integration:** The application does not connect to banking systems, card readers, UPI services, or payment gateways. Payment is entered manually for simulation.
5. **No Hardware Integration:** The program does not communicate with fuel dispensers, flow meters, storage tanks, or other petrol pump equipment.
6. **Limited Reporting:** It does not generate daily sales reports, monthly summaries, inventory reports, or cash reconciliation statements.
7. **Limited Security:** The current version does not include staff authentication, user roles, or access control.
8. **Limited Accounting Features:** The system does not implement comprehensive taxation, accounting, or financial reconciliation procedures.
9. **Basic Input Validation:** It handles common invalid inputs, but it does not provide the advanced validation, error recovery, and audit controls expected in a commercial application.
10. **Educational Purpose Only:** The system is intended for learning and demonstration. It cannot replace certified fuel-dispensing equipment, approved point-of-sale systems, or applicable safety and regulatory procedures.

Impact Analysis

1. Impact of Project on Society

a. Positive Impact of the Project on Society:

- Reduces manual calculation errors in billing.
- Saves time by automating fuel-cost calculations.
- Improves transparency by displaying clear bill details.
- Promotes the use of computer-based billing systems.
- Helps students learn practical C++ programming concepts.

b. Negative Impact of the Project on Society:

- Incorrect fuel prices or quantities may result in incorrect bills.
- The system cannot connect to actual fuel dispensers.
- Lack of database and security features limits practical use.
- Using the simulation for real transactions without proper validation may cause billing errors.
- The system requires further development to support complete petrol pump operations.

2. Impact of Project on Environment

a. Positive Impact of Project on Environment:

- Reduces paper usage by demonstrating computerized billing.
- Minimizes paper waste through digital processing.
- Promotes the use of software for efficient business operations.
- Provides a foundation for future paperless billing systems.

b. Negative Impact of Project on Environment:

- Computers consume electricity during operation.
- Electronic devices may contribute to e-waste when discarded improperly.
- Printing receipts in future versions may increase paper consumption.
- Increased use of electronic equipment may contribute indirectly to carbon emissions.

Professional Ethical Practices To Be Followed

1. Accuracy and Reliability: Ensure that fuel prices, quantities, and billing calculations are accurate.

2. Honesty and Transparency: Display correct billing information and avoid misleading users.

3. Data Privacy: Protect customer and transaction information if storage features are added.

4. Software Quality: Follow proper coding standards, testing procedures, and input validation practices.

5. Safety and Responsibility: Do not connect the educational simulation to real fuel dispensers without certified equipment and proper authorization.

6. Legal Compliance: Follow applicable rules and regulations when developing any real-world petrol pump billing system.

7. Professional Competence: Continuously improve technical skills and maintain the software responsibly.

8. Environmental Responsibility: Encourage paperless billing and efficient use of computer resources.

Industry Relevance

The Indian Petrol Pump Billing System Using C++ is relevant to companies involved in fuel retailing, petroleum distribution, and automated billing solutions. The project demonstrates basic features such as fuel selection, price calculation, payment handling, and receipt generation, which are also components of larger commercial fuel-management systems.

1. Indian Oil Corporation Limited (IOCL)

Relevance to Project: The project aligns with IOCL's petrol pump operations by demonstrating fuel billing, price calculation, and customer transaction processing.

Application Summary: IOCL operates fuel retail outlets that sell petrol and diesel. A billing system like this project's basic model can support fuel-cost calculations and receipt generation within a larger retail automation system.

Website: Official Website

2. Bharat Petroleum Corporation Limited (BPCL)

Relevance to Project: The project shares similarities with BPCL's digital fuelling solutions, particularly in billing transparency and payment processing.

Application Summary: BPCL's UFill service enables customers to manage fuel purchases through digital technology, QR codes, and supported payment applications, improving convenience and transaction transparency.



Official Website of BPCL, India

+1

Website: Official Website

3. Hindustan Petroleum Corporation Limited (HPCL)

Relevance to Project: The project demonstrates basic billing functions similar to those used within HPCL's more advanced petrol pump automation systems.

Application Summary: HPCL's retail automation systems support transaction recording, fuel-price updates, sales monitoring, inventory management, and integrated digital payments across its automated fuel stations.



Hindustan Petroleum Corporation Ltd.

+1

Website: Official Website

4. Gilbarco Veeder-Root

Relevance to Project: The project uses similar core concepts of fuel billing and transaction calculation that are incorporated into commercial fuel-station point-of-sale systems.

Application Summary: Gilbarco Veeder-Root provides fuel dispensers and point-of-sale solutions that support fuel sales, payment processing, sales reconciliation, and fuel-station management.



Gilbarco Veeder-Root - India

+1

Website: Official Website

5. Nayara Energy

Relevance to Project: The project is relevant to Nayara Energy's retail fuel business because it demonstrates the basic process of calculating fuel bills and handling customer payments.

Application Summary: Nayara Energy operates a network of fuel stations offering petrol, diesel, and multiple payment options. Its retail operations provide a practical context for billing applications and customer transaction management.



Nayara Energy

Website: Official Website

Project Planning (Timeline Analysis and Finance Management

Week	Planned Activities	Expected Outcome
Week 1	Topic selection, requirement analysis, and planning the system workflow.	Finalized requirements and project design.
Week 2	Develop the algorithm, flowchart, and C++ code for fuel selection and bill calculation.	Basic billing system implemented.
Week 3	Implement payment calculation, receipt generation, input validation, and testing.	Functional and tested program.
Week 4	Debugging, preparing sample outputs, report writing, and final submission.	Completed project report and demonstration.

Resource	Estimated Cost (₹)
Computer/Laptop (existing)	0
C++ Compiler and IDE (free)	0
Internet and electricity	150
Report printing and binding	200
Miscellaneous expenses	100
Total Estimated Cost	450

Future Scope and Further Enhancement

The Indian Petrol Pump Billing System is currently a basic C++ console application developed for educational purposes. In the future, it can be enhanced into a complete petrol pump management system by adding advanced features, improving security, and integrating with modern technologies.

Future Scope

1**. Database Integration**: Store customer transactions, fuel sales, and payment records using MySQL or SQLite for better record management.

2. Graphical User Interface (GUI): Replace the console interface with a user-friendly GUI using Qt, Java, or C# to improve usability.

3. Real-Time Fuel Price Updates: Automatically update petrol and diesel prices from authorized sources instead of using fixed values.

4. Digital Payment Integration: Support UPI, debit/credit cards, and QR-code payments for faster and more convenient transactions.

5. Staff Login and Security: Add secure login for employees and administrators with role-based access control.

6. Sales Reports and Analytics: Generate daily, weekly, and monthly sales reports, fuel consumption statistics, and revenue summaries.

7. Transaction History: Save receipts with unique transaction IDs, timestamps, and customer billing history.

8. Inventory Management: Monitor fuel stock levels and provide alerts when fuel quantity reaches a minimum threshold.

9. Receipt Printing and Email Bills: Enable receipt printing and digital receipt sharing through SMS or email.

10. Cloud-Based Management: Connect multiple petrol pumps to a cloud platform for centralized monitoring and management.

Further Enhancement

- Integrate with real fuel dispensers using certified hardware interfaces.
- Add barcode or RFID-based vehicle identification for fleet customers.
- Include GST and accounting features for commercial use.
- Develop a mobile application for customer services and digital receipts.
- Implement AI-based sales forecasting and inventory prediction.

These enhancements can transform the project from a simple educational billing simulation into a modern, secure, and efficient petrol pump management system suitable for real-world business applications.

PROGRAM

```
#include <iostream>
#include <iomanip>
#include <string>
#include <limits>
using namespace std;

struct Fuel {
   string name;
   double pricePerLitre;
};

void clearBadInput() {
   cin.clear();
   cin.ignore(numeric_limits<streamsize>::max(), '\n');
}

int main() {
   // Sample prices for educational purposes only
   Fuel fuels[2] = {
      {"Petrol", 104.50},
      {"Diesel", 92.50}
   };

   char again = 'y';

   cout << fixed << setprecision(2);

   do {
      int choice;
      double litres, payment;

      cout << "\n================================\n";
      cout << " INDIAN PETROL PUMP BILLING SYSTEM\n";
      cout << "================================\n";
      cout << "1. Petrol\n";
      cout << "2. Diesel\n";
      cout << "Choose fuel (1 or 2): ";

      if (!(cin >> choice) || choice < 1 || choice > 2) {
         cout << "Invalid fuel selection!\n";
         clearBadInput();
         continue;
      }

      cout << "Enter quantity in litres: ";
```

```
        if (!(cin >> litres) || litres <= 0) {
            cout << "Please enter a valid positive quantity.\n";
            clearBadInput();
            continue;
        }

        Fuel selected = fuels[choice - 1];
        double total = litres * selected.pricePerLitre;

        cout << "\n---------- BILL ----------\n";
        cout << "Fuel Type      : " << selected.name << '\n';
        cout << "Price per Litre : Rs. "
             << selected.pricePerLitre << '\n';
        cout << "Quantity       : " << litres << " L\n";
        cout << "Total Amount    : Rs. " << total << '\n';

        cout << "Enter payment amount: Rs. ";

        if (!(cin >> payment) || payment < 0) {
            cout << "Invalid payment amount!\n";
            clearBadInput();
            continue;
        }

        if (payment < total) {
            cout << "Balance Due     : Rs. "
                 << total - payment << '\n';
        } else {
            cout << "Change Returned : Rs. "
                 << payment - total << '\n';
        }

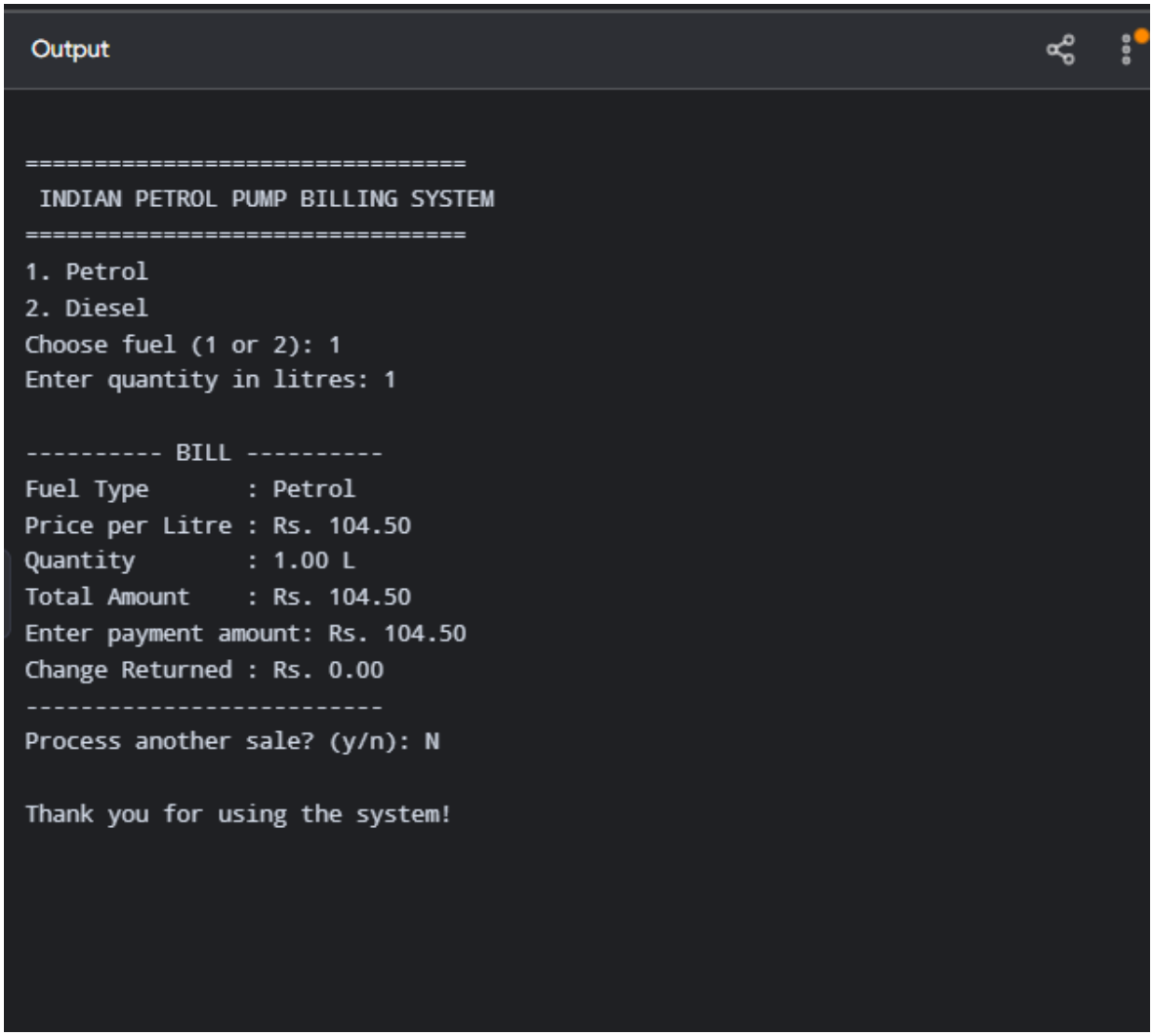
        cout << "--------------------------\n";
        cout << "Process another sale? (y/n): ";
        cin >> again;
        clearBadInput();

    } while (again == 'y' || again == 'Y');

    cout << "\nThank you for using the system!\n";

    return 0;
}
```

Expected Outcome



References

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[4] Simon Haykin, "Cognitive Dynamic Systems", *Proceedings of the IEEE* , Vol. 94, No. 11, November 2006, 1910-1911.